

Module: `tf.compat.v1.math` Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/math

Public API for `tf._api.v2.math` namespace

Function Signatures

```
abs(...)  
accumulate_n(...)  
acos(...)  
acosh(...)
```

Documentation

Public API for `tf._api.v2.math` namespace

Modules

special module: Public API for `tf._api.v2.math.special` namespace

Functions

`abs(...)`: Computes the absolute value of a tensor.

`accumulate_n(...)`: Returns the element-wise sum of a list of tensors.
(deprecated)

`acos(...)`: Computes $\arccos$ of x element-wise.

`acosh(...)`: Computes inverse hyperbolic cosine of x element-wise.

`add(...)`: Returns $x + y$ element-wise.

`add_n(...)`: Returns the element-wise sum of a list of tensors.

`angle(...)`: Returns the element-wise argument of a complex (or real) tensor.

`approx_max_k(...)`: Returns max k values and their indices of the input operand in an approximate manner.

`approx_min_k(...)`: Returns min k values and their indices of the input operand in an approximate manner.

`argmax(...)`: Returns the index with the largest value across axes of a tensor.
(deprecated arguments)

`argmin(...)`: Returns the index with the smallest value across axes of a tensor.
(deprecated arguments)

`asin(...)`: Computes the trigonometric inverse sine of x element-wise.

`asinh(...)`: Computes inverse hyperbolic sine of x element-wise.

`atan(...)`: Computes the trigonometric inverse tangent of x element-wise.

`atan2(...)`: Computes arctangent of y/x element-wise, respecting signs of the arguments.

`atanh(...)`: Computes inverse hyperbolic tangent of x element-wise.

`bessel_i0(...)`: Computes the Bessel i_0 function of x element-wise.

`bessel_i0e(...)`: Computes the Bessel i_0e function of x element-wise.

bessel_i1(...): Computes the Bessel i1 function of x element-wise.

bessel_i1e(...): Computes the Bessel i1e function of x element-wise.

betainc(...): Compute the regularized incomplete beta integral $I_x(a, b)$.

bincount(...): Counts the number of occurrences of each value in an integer array.

ceil(...): Return the ceiling of the input, element-wise.

confusion_matrix(...): Computes the confusion matrix from predictions and labels.

conj(...): Returns the complex conjugate of a complex number.

cos(...): Computes cos of x element-wise.

cosh(...): Computes hyperbolic cosine of x element-wise.

count_nonzero(...): Computes number of nonzero elements across dimensions of a tensor. (deprecated arguments) (deprecated arguments)

cumprod(...): Compute the cumulative product of the tensor x along axis.

cumsum(...): Compute the cumulative sum of the tensor x along axis.

cumulative_logsumexp(...): Compute the cumulative log-sum-exp of the tensor x along axis.

digamma(...): Computes Psi, the derivative of Lgamma (the log of the absolute value of

divide(...): Computes Python style division of x by y.

divide_no_nan(...): Computes a safe divide which returns 0 if y (denominator) is zero.

equal(...): Returns the truth value of $(x == y)$ element-wise.

erf(...): Computes the Gauss error function of x element-wise. In statistics, for non-negative values of x , the error function has the following interpretation: for

a random variable Y that is normally distributed with mean 0 and variance $1/\sqrt{2}$, $\text{erf}(x)$ is the probability that Y falls in the range $[-x, x]$.

`erfc(...)`: Computes the complementary error function of x element-wise.

`erfcinv(...)`: Computes the inverse of complementary error function.

`erfinv(...)`: Compute inverse error function.

`exp(...)`: Computes exponential of x element-wise. $y = e^x$.

`expm1(...)`: Computes $\exp(x) - 1$ element-wise.

`floor(...)`: Returns element-wise largest integer not greater than x .

`floordiv(...)`: Divides x / y elementwise, rounding toward the most negative integer.

`floormod(...)`: Returns element-wise remainder of division.

`greater(...)`: Returns the truth value of $(x > y)$ element-wise.

`greater_equal(...)`: Returns the truth value of $(x \geq y)$ element-wise.

`igamma(...)`: Compute the lower regularized incomplete Gamma function $P(a, x)$.

`igammac(...)`: Compute the upper regularized incomplete Gamma function $Q(a, x)$.

`imag(...)`: Returns the imaginary part of a complex (or real) tensor.

`in_top_k(...)`: Says whether the targets are in the top K predictions.

`invert_permutation(...)`: Computes the inverse permutation of a tensor.

`is_finite(...)`: Returns which elements of x are finite.

`is_inf(...)`: Returns which elements of x are Inf.

`is_nan(...)`: Returns which elements of x are NaN.

`is_non_decreasing(...)`: Returns True if x is non-decreasing.

`is_strictly_increasing(...)`: Returns True if x is strictly increasing.

`l2_normalize(...)`: Normalizes along dimension axis using an L2 norm.
(deprecated arguments)

`lbeta(...)`: Computes $\ln(|\text{Beta}(x)|)$, reducing along the last dimension.

`less(...)`: Returns the truth value of $(x < y)$ element-wise.

`less_equal(...)`: Returns the truth value of $(x \leq y)$ element-wise.

`lgamma(...)`: Computes the log of the absolute value of Gamma(x) element-wise.

`log(...)`: Computes natural logarithm of x element-wise.

`log1p(...)`: Computes natural logarithm of $(1 + x)$ element-wise.

`log_sigmoid(...)`: Computes log sigmoid of x element-wise.

`log_softmax(...)`: Computes log softmax activations. (deprecated arguments)

`logical_and(...)`: Returns the truth value of x AND y element-wise.

`logical_not(...)`: Returns the truth value of NOT x element-wise.

`logical_or(...)`: Returns the truth value of x OR y element-wise.

`logical_xor(...)`: Logical XOR function.

`maximum(...)`: Returns the max of x and y (i.e. $x > y ? x : y$) element-wise.

`minimum(...)`: Returns the min of x and y (i.e. $x < y ? x : y$) element-wise.

`mod(...)`: Returns element-wise remainder of division.

`multiply(...)`: Returns an element-wise $x * y$.

`multiply_no_nan(...)`: Computes the product of x and y and returns 0 if the y is zero, even if x is NaN or infinite.

`ndtri(...)`: Compute quantile of Standard Normal.

`negative(...)`: Computes numerical negative value element-wise.

`nextafter(...)`: Returns the next representable value of `x1` in the direction of `x2`, element-wise.

`not_equal(...)`: Returns the truth value of `(x != y)` element-wise.

`polygamma(...)`: Compute the polygamma function $\psi^{(n)}(x)$.